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3 **NetGO 3.0: Protein Language Model Improves Large-scale** 4 **Functional Annotations**

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Abstract

As one of the state-of-the-art automated function prediction (AFP) methods, NetGO 2.0 integrates multi-source information to improve the performance. However, it mainly utilizes the proteins with experimentally supported functional annotations without leveraging valuable information from a vast number of unannotated proteins. Recently, protein language models have been proposed to learn informative representations (e.g., Evolutionary Scale Modelling (ESM)-1b embedding) from protein sequences based on self-supervision. We represent each protein by ESM-1b and use logistic regression (LR) to train a new model, LR-ESM, for AFP. The experimental results show that LR-ESM achieves comparable performance with the best-performing component of NetGO 2.0. Therefore, by incorporating LR-ESM into NetGO 2.0, we develop NetGO 3.0 to improve the performance of AFP extensively. NetGO 3.0 is freely accessible at <https://dmiip.sjtu.edu.cn/ng3.0>.

KEYWORDS: Protein function prediction; Web service; Protein language models

Introduction

Proteins are complex molecules that play essential roles in various biological activities. To understand the underlying mechanism of an organism as a physical system, annotating the functions of proteins is a crucial task. Gene Ontology came into being in 1998 to describe varying levels of functional information on gene/RNA/protein, which contains three domains: Molecular Function Ontology (MFO), Biological Process Ontology (BPO), and Cellular Component Ontology (CCO) with over 40 000 terms [1]. As of Nov 2022, the number of raw protein sequences is more than 230 million in UniProtKB, but less than 0.1% of them have experimental annotations [2]. It is thus desirable to develop high-performing computational methods to achieve automated function prediction without costly experiments [3].

AFP is a large-scale multi-label classification problem where multiple related GO terms are assigned to a target protein. In the last few years, several high-performing web servers have been developed for AFP, such as INGA 2.0 [4], DeepGOWeb [5], MetaGO [6], and QAUST [7]. Under the learning to rank (LTR) framework [8], GOLabeler [9], NetGO [10], and NetGO 2.0 [11] achieved a state-of-the-art performance in the recent community-wide Critical Assessment of Functional Annotation (CAFA) [3]. Specifically, NetGO 2.0 integrates protein information from different sources to encode proteins in a computer-understandable way, such as sequences, protein domains, protein-protein interaction networks, and scientific literature. However, it does not leverage valuable information from unannotated proteins (>99.9% of all known proteins).

Recently, the idea of pre-training in natural language processing [12] has been extended to build protein language models using self-supervised learning on millions of sequences [13-15]. Most protein language models predict the masked or next amino acid within a sequence and generate protein embeddings that can generalize across downstream tasks (more details shown in Supplementary Materials). Some recent studies have explored protein language models for AFP [16, 17]. However, they have a common limitation: less frequent GO terms (e.g., having less than 40 annotated

proteins) are excluded in the evaluation, which accounts for around 75% of total annotations in the CAFA setting. In this work, we predict the associations between proteins and each GO term based on ESM-1b embeddings, which were trained on over 250 million protein sequences [13]. Our experimental results show that the learned representations are helpful to AFP. Therefore, we develop NetGO 3.0 by incorporating ESM-1b embeddings in order to improve the performance extensively, which highlights the predictive power of the protein language model for AFP.

Materials and methods

Protein language models

A challenging problem is figuring out how to represent protein sequences as fixed-length vectors that capture the realistic sequence-function relationship. Traditional methods rely on a holistic understanding of protein properties. Recently, protein language models have provided a solution that interprets protein sequences as sentences and amino acids as words to extract fundamental features of a protein with rich and systematic information. Protein language models train nonlinear neural networks with an unsupervised objective on a large-scale dataset of protein sequences [13-15, 21].

Generally, protein language models apply deep learning models such as recurrent neural networks (RNN) and Transformer to achieve statistical embeddings of proteins from tremendous sequences. UniRep represented protein sequences as fixed-length vectors by LSTM with ~24 million sequences [15]. TAPE distilled protein properties from sequences by semi-supervised learning based on ResNet, LSTM, and Transformer; then evaluated their performance on five biologically relevant tasks [21]. Moreover, a multi-task learning framework was recently proposed to incorporate structural information (e.g., contact maps and structural similarity prediction tasks) to enrich protein language models [22]. Furthermore, researchers applied protein language models to study protein molecular function prediction[17]. UDSMProt put forward a task-agnostic representation for proteins and achieved good performance on protein-

level prediction tasks, namely enzyme class prediction and GO prediction [16]. However, both methods should have considered less frequent GO terms.

In this paper, a new component LR-ESM in NetGO 3.0 is proposed to utilize ESM-1b, a 34-layer Transformer-based model trained on UniPAC database with 250M protein sequences and 650M parameters, to generate protein-level representations by average pooling across all residue-level embeddings [13].

Implementation

NetGO 2.0 integrates seven component methods, which are Naive, BLAST-KNN, LR-3mer, LR-InterPro, NetKNN, LR-Text, and Seq-RNN. We replace Seq-RNN with LR-ESM in NetGO 3.0, which makes function prediction based on a protein language model. Specifically, LR-ESM utilizes ESM-1b, a 34-layer Transformer-based model trained on the UniPac database with 250 million sequences [13], to generate protein embeddings and complete prediction. As ESM-1b has a limitation of protein sequence length, we keep the first 1000 amino acids for those protein sequences longer than 1024. We then use ESM-1b to encode each amino acid as an embedding of size 1280 for a target protein. To obtain the protein-level embedding, we apply the operation of average pooling on all amino acid positions, which comprehensively collects information from sequence data alone. Finally, LR-ESM utilizes protein embeddings as input to train logistic regression classifiers and estimates the association between target proteins and each GO term.

Benchmark datasets

As NetGO 2.0 collected the data following the setting of CAFA, we utilized the same benchmark dataset to evaluate the performance of NetGO 3.0 and the competing methods. Table S1 in supplementary materials reports the number of proteins in the benchmark dataset.

To take advantage of the latest annotation data, we collect sequences and GO terms before January 2022 from UniProt [2], GOA [23] and GO [1]. Similarly, we train and update our model on the new dataset by following the standard protocols of NetGO 2.0 [11].

1. Training data: all experimental annotation data before January 2020.

2. Validation data: all experimental *no-knowledge* and *limited-knowledge* proteins annotated from January 2020 to December 2020.

3. Testing data: all experimental *no-knowledge* proteins between January 2021 and December 2021.

More details for the new dataset are listed in Section S2 and Table S2 of supplementary materials.

Results

We compare the performance of NetGO 3.0 with the competing methods on the benchmark dataset from NetGO 2.0. The performance is evaluated by area under the precision-recall curve (AUPR) and two standard metrics in CAFA, F_{max} and S_{min} .

The definitions of three metrics are given in Section S1 of supplementary materials.

Performance comparisons of NetGO 3.0 with its component methods and competing methods

Table 1 illustrates the test results for NetGO 3.0, NetGO 2.0, GOLabeler, DeepGOWeb, and the component methods of NetGO 3.0. GOLabeler and NetGO 2.0 achieved top performance in CAFA3 and CAFA4, respectively [9, 11]. DeepGOWeb provided an accurate prediction for protein function by deep learning [5]. The underlined numbers in the upper table imply the best performance for component methods.

We select Naive, BLAST-KNN, and Seq-RNN [11] from NetGO 2.0 as three baseline methods. The Naive method annotates each pair of protein and GO term with a score that equals the probability of the term appearing in the training data. BLAST-KNN assigns a protein with GO terms based on annotations of its top BLAST hits [9]. While the first two are component methods inherited from both NetGO and NetGO2.0, Seq-RNN is a new component of NetGO2.0, which was designed to extract the deep representation of a protein sequence [11]. As shown in **Figure 1**, LR-ESM outperforms baseline methods on all three GO domains. As a replacement for Seq-RNN, LR-ESM achieves a better performance. Specifically, in terms of F_{max} , LR-ESM achieves 21.6%, 31.3%, and 7.5% improvements over Seq-RNN on MFO, BPO, and CCO, respectively,

which indicates the effectiveness of ESM-1b for AFP. Moreover, LR-ESM and LR-InterPro show comparable performance in all three GO domains. Note that, in terms of S_{min} , LR-ESM outperforms all other component methods and even achieves a better performance on MFO than NetGO 2.0. Therefore, it is reasonable to construct a more robust model by incorporating LR-ESM into NetGO 2.0.

Furthermore, we compare NetGO 3.0 with several high-performing methods in CAFA. In the lower part of Table 1, we highlight the best performance in bold among these methods. The experimental results show that NetGO 3.0 has achieved a more superior performance than the competing methods. In terms of F_{max} and S_{min} , NetGO 3.0 achieves a better performance in all three GO domains. For example, NetGO 3.0 achieved the highest F_{max} of 0.378 in BPO, which was 3.3% and 16.0% improvements over NetGO 2.0 (0.366) and GOLabeler (0.326), respectively. Finally, we calculated the confidence interval (95%) for each method by bootstrapping with 100 iterations on the test set, as shown in Figure 1. The results demonstrate that NetGO 3.0 can benefit from protein language models with deep dense embeddings.

To better illustrate the strength of NetGO 3.0, we draw a Venn diagram in **Figure 2** to show the overlaps and differences among the prediction results of GOLabeler, DeepGOWeb, and NetGO 3.0. There are three main findings: (i) Although each method can predict distinct GO terms, the prediction results of the three methods are overlapped substantially, especially in CCO. Specifically, there are 6.96 GO terms assigned to one protein on average that are predicted by all three methods in CCO, which accounts for 62.5%, 70.1%, and 77.3% in the prediction results of DeepGOWeb (11.14), GOLabeler (9.84), and NetGO 3.0 (9.00), respectively. (ii) DeepGOWeb predicted more GO terms but achieved lower performance than the other two methods, indicating that false positive GO terms are common in the prediction results. For example, DeepGOWeb predicted 21.34 distinct GO terms and achieved the lowest F_{max} of 0.305 in BPO, which suggests that most of its predicted GO terms are incorrect. (iii) Compared with MFO and CCO, NetGO 3.0 and GOLabeler differs significantly in predicting GO terms

in BPO. Specifically, in terms of BPO, although the 15.42 GO terms predicted by the two methods are consistent, the number of distinct GO terms predicted by NetGO 3.0 and GOLabeler are 9.59 and 8.18, respectively. We note that NetGO 3.0 performed better than GOLabeler in BPO in terms of F_{max} , where NetGO 3.0 (0.378) achieved a 16.0% improvement compared to GOLabeler (0.326). It demonstrates that NetGO 3.0 is more accurate and can predict more true positive terms for query proteins.

Performance on specific species (HUMAN and MOUSE)

Species-specific analyses are helpful for researchers to study a certain species. Here, we explore the performance of different AFP methods over two model species, HUMAN and MOUSE. **Tables 2** and **3** show the performance of NetGO 3.0 and the competing methods for proteins in HUMAN and MOUSE. According to the performance of the two species, we can see that all methods obtain a better prediction performance on HUMAN in comparison with MOUSE. For example, LR-InterPro, LR-ESM, and NetGO 3.0 achieved higher AUPRs of 0.704, 0.690, and 0.730 over HUMAN proteins in MFO, whereas the three methods only achieved AUPRs of 0.601, 0.615, and 0.620 over MOUSE proteins. The annotation information for different species is from different databases, which may lead to the difference. Moreover, LR-ESM again achieves a similar performance as LR-InterPro in both species, which strongly demonstrates that features extracted by ESM-1b are as robust as InterProScan among many species.

For HUMAN and MOUSE, NetGO 3.0 outperforms all competing methods in all three GO domains. Specifically, NetGO 3.0 performs better than NetGO 2.0 in HUMAN BPO prediction, which achieves 9.3% and 9.5% improvements in terms of F_{max} and AUPR. Further, the results highlight the importance of source data and the effectiveness of the protein language model.

Performance comparisons over groups categorized by the number of annotations per GO term

We divided GO terms in the test dataset into three groups according to the number of annotations per GO term: 10-30, 31-100, and >100. **Table 4** reports the M-AUPR

computed in each group, where M-AUPR is GO term-centric by averaging AUPR on each GO term. LR-ESM outperforms other component methods in most cases, which indicates that ESM-1b embeddings are informative. Note that LR-ESM consistently ranks higher than LR-InterPro for three domains in the first group, especially for BPO, which obtains a 47.8% improvement. It proves that protein embeddings are effective with such a vast amount of training data for AFP.

NetGO 3.0 achieves the best results among all the methods in every group and domain, and the improvement over NetGO 2.0 is especially significant in CCO. Specifically, the advances made by NetGO 3.0 are 28.7%, 8.4%, and 8.2% for the three groups, respectively. Moreover, we collect the CCO terms in the second and third layers annotated with more than ten proteins in the test set. As shown in **Figure 3**, NetGO 3.0 achieves a better performance on most GO terms, which strongly suggests that ESM-1b is powerful for predicting protein functions about cellular components.

Performance comparison on difficult proteins

Following the CAFA setting, proteins with a BLAST identity of less than 0.6 to any protein in training data are identified as “difficult proteins” [3]. In the test set, there are 66, 85, and 70 difficult proteins in MFO, BPO, and CCO, respectively. It is evident that methods based on homology find it hard to predict the function of difficult proteins accurately. **Table 5** shows the performance of different methods in dealing with difficult proteins. As mentioned above, BLAST-KNN, a method that annotated target proteins by homology proteins, ranked last in 9 experimental settings. We find that LR-InterPro and LR-ESM are the two best-performing component methods in this scenario. For example, in terms of S_{min} , there is a slight difference between the two methods in three domains. LR-ESM and LR-InterPro achieve the best performance for all component methods in 6 and 3 out of 9 settings. Once again, NetGO 3.0 is proved to be the best method for predicting the function of difficult proteins.

Performance comparisons on proteins with sequence lengths longer than 1000

We performed a truncation operation for proteins longer than 1000 so that ESM-1b could generate representations for all proteins in the dataset. Focusing on the

performance of each method on these long proteins helps us better understand the advantages and limitations of NetGO 3.0. There exist 21, 78, and 26 test proteins in MFO, BPO, and CCO. **Table 6** lists the prediction results of component methods, NetGO 2.0 and NetGO 3.0. As shown in Table 6, LR-ESM is no longer one of the best-performing component methods, which indirectly causes NetGO 3.0 to perform worse than NetGO 2.0 in MFO and BPO. By comparing each method's performance on the entire test set in Table 1, we noticed that the performance decreased for all methods except Net-KNN. This suggests that function prediction for long proteins is a challenge.

Moreover, we observed the prediction performance of NetGO 2.0 and NetGO 3.0 on several unannotated proteins Q3UZV7, F1QKQ1, and Q2HX28. The sequence lengths of these three proteins are 1028, 1356, and 1409. As shown in **Table 7**, NetGO 2.0 achieved better AUPR on three proteins, which indicates that the truncated sequences in long proteins are important sources of information and are critical for predicting functions. This further confirms that NetGO 3.0 needs to be improved in handling long sequences, which will be important future research work.

Visualization of the predicted results

We present more options to visualize predicted GO terms to better illustrate prediction results. Compared to NetGO 2.0, the new web server offers a novel perspective to present the results, which can provide more relevant information about predicted GO terms. **Figure 4** shows the new result page of NetGO 3.0, which mainly includes three ways to visualize the prediction performance. Although GO terms in top layers usually achieve a higher score and rank higher, NetGO 3.0 clarifies the depth of predicted GO terms, which allows users to find specific GO terms in bottom layers. Note that the colour in the result page and node size in Figure 4.D are determined by the predicted confidence score, which can help users better understand the predicted results in an original view.

Case study

Finally, we select a specific protein as input and show the results obtained by NetGO 3.0 and its competing methods. Ubiquitin-like protein 5 (UniProt ID: Q9FGZ9) is a difficult protein with low BLAST similarity to training proteins. **Table 8** shows the 18

GO terms in BPO annotated to protein Q9FGZ9. **Figure 5** also depicts the directed acyclic graphs (DAG) according to the relationship of 18 GO terms in Gene Ontology. In Table 8, it is clear that BLAST-KNN failed to achieve a valid result because homology-based methods were not suitable for difficult protein function prediction. LR-InterPro and LR-ESM extracted features from raw amino acid sequences and obtained better results than BLAST-KNN. In the top 20 predicted GO terms, the number of true positive samples achieved by LR-ESM is significantly larger than other methods, which predicted 14 correct function labels. NetGO and NetGO 2.0 predicted only six correct GO terms, which are not competitive compared to LR-ESM and NetGO 3.0. The reason for this phenomenon may be that the new component method, LR-ESM, is more robust for difficult proteins than other methods and is able to represent them more efficiently. With the support of the protein language model, NetGO 3.0 achieved 15 true GO terms out of 19 predicted ones, which successfully predicted the GO terms that NetGO and NetGO 2.0 failed to predict. Figure 5 illustrates the hierarchy of correct and predicted GO terms, indicating that NetGO 3.0 is able to predict those GO terms with less information in the deeper layers. Overall, this typical example demonstrates that the high predictive performance of NetGO 3.0 is closely related to the protein language models.

Conclusion

We have developed NetGO 3.0 to improve the performance of large-scale AFP by incorporating a new component LR-ESM, which utilizes a protein language model to generate powerful representations of proteins. Interesting future work would be integrating protein structural information into NetGO to enhance the performance of AFP [18-20].

Availability

The web server of NetGO 3.0 is freely accessible at <https://dmiip.sjtu.edu.cn/ng3.0>.

CRediT author statement

Shaojun Wang: Data Curation, Software, Methodology, Writing - original draft.
Ronghui You: Conceptualization, Methodology, Writing - review & editing. **Yunjia Liu:** Methodology, Visualization, Writing - review & editing. **Yi Xiong:** Resource, Writing - review & editing. **Shanfeng Zhu:** Conceptualization, Resource, Writing – review & editing. All authors read and approved the final manuscript.

Competing interests

The authors have declared no competing interests.

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Figure legends

Figure 1 The performance comparison on F_{max} and S_{min}

The performance of Naive, BLAST-KNN, SeqRNN, LR-ESM, NetGO 2.0, and NetGO 3.0 on the benchmark dataset of NetGO 2.0 over three GO domains. The arrows imply higher values for F_{max} and lower for S_{min} over three GO domains. The bars denote the confidence intervals (95%) calculated by bootstrapping with 100 iterations as in CAFA.

Figure 2 The overlap and difference among the predicted GO terms of GOLabeler, DeepGOWeb, and NetGO 3.0

The Venn diagram depicts the overlap and difference among the predicted GO terms of GOLabeler, DeepGOWeb, and NetGO 3.0. The figures in the graph represent the average number of predicted GO terms over test proteins in three methods.

Figure 3 The performance over M-AUPR in CCO

There are 31 GO terms in the second and third layers of the CCO domain that are annotated with more than ten proteins in the test set. By checking the performance improvement of NetGO 3.0 over 2.0 on each term, the GO terms that have more than a 5% increase or decrease are shown in red and green, respectively.

Figure 4 Visualization of prediction results on the web server

A. Prediction result page of NetGO 3.0's website. "GO DAG", "Bar plot" and "Bubble plot" are the new interfaces to visualize the predicted GO terms. We also added a new column named "Depth" to show the depth of GO terms in Gene Ontology. B. The predicted GO terms and their directed acyclic graph. C. We plot the predicted GO terms and their confidence scores in a bar diagram. D. The Predicted GO term and its depth in Gene Ontology are plotted in a bubble diagram.

Figure 5 DAG of GO terms associated with Q9FGZ9 in BPO

Each GO term is attached with tags, which illustrates that the GO term is predicted correctly by corresponding methods.

Tables

Table 1 Performance of NetGO 3.0 with its components and competing methods on the test set

Table 2 Performance comparisons for proteins in HUMAN

Table 3 Performance comparisons for proteins in MOUSE

Table 4 Performance comparisons over groups categorized by the number of annotations per GO term

Table 5 Performance on difficult proteins

Table 6 Performance comparisons on proteins with sequence lengths longer than 1000

Table 7 Performance on three long proteins in BPO

Table 8 Predicted GO terms of Q9FGZ9 in BPO by NetGO 3.0 and competing methods

Each method shows the top 20 predicted GO terms (the root term GO:0008150 is deleted). Correctly predicted GO terms are in red, and the last row shows the groundtruth GO terms.

Supplementary material

Section S1 Definition of performance evaluation metrics

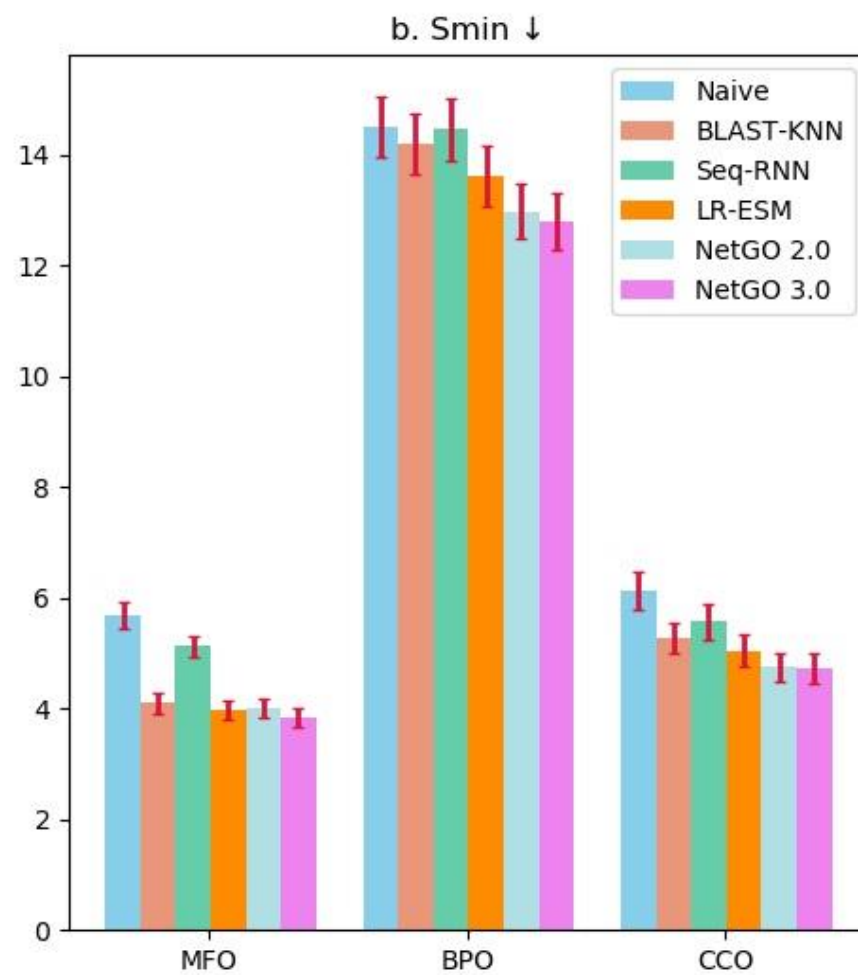
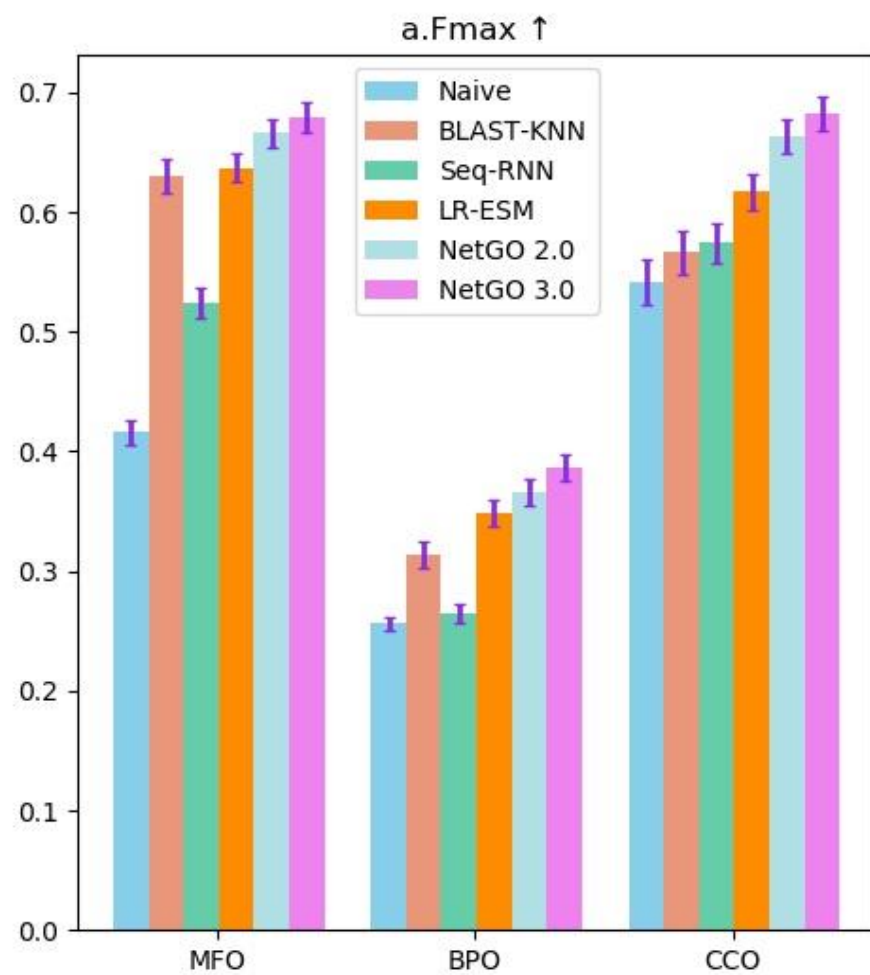
Section S2 Details of the new dataset

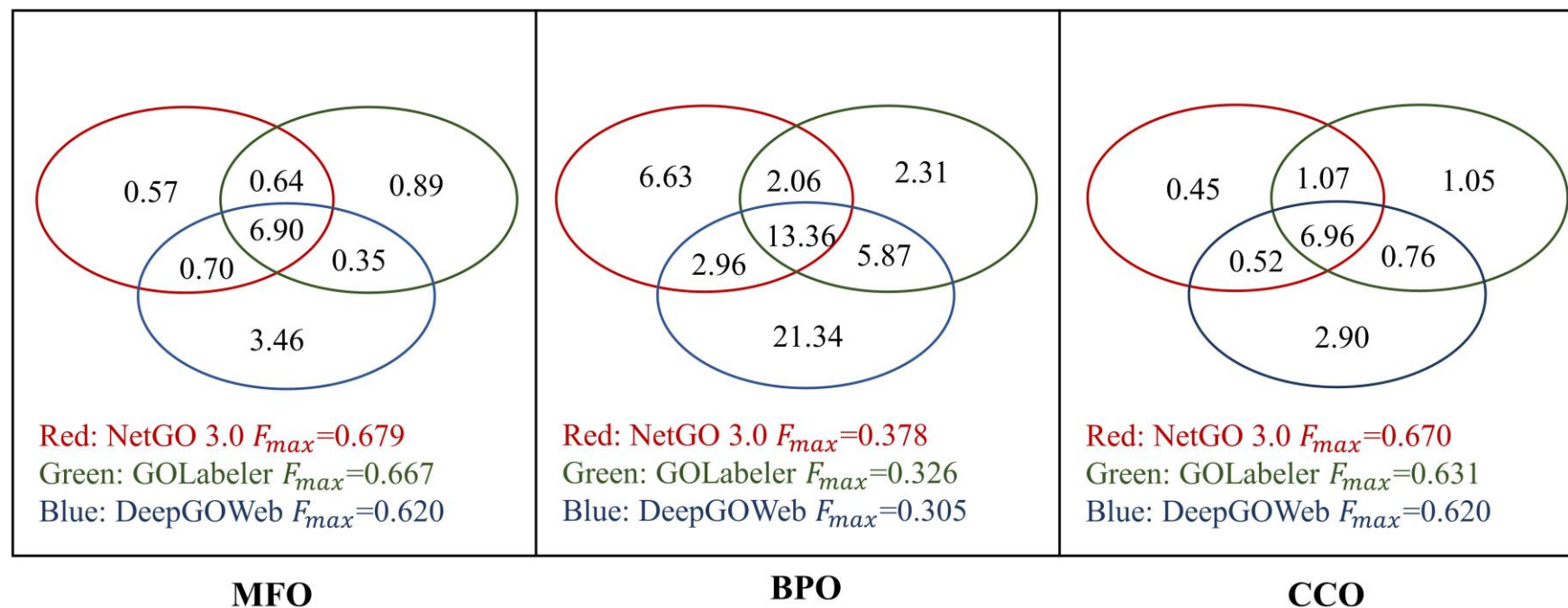
Section S3 Performance of different competing models

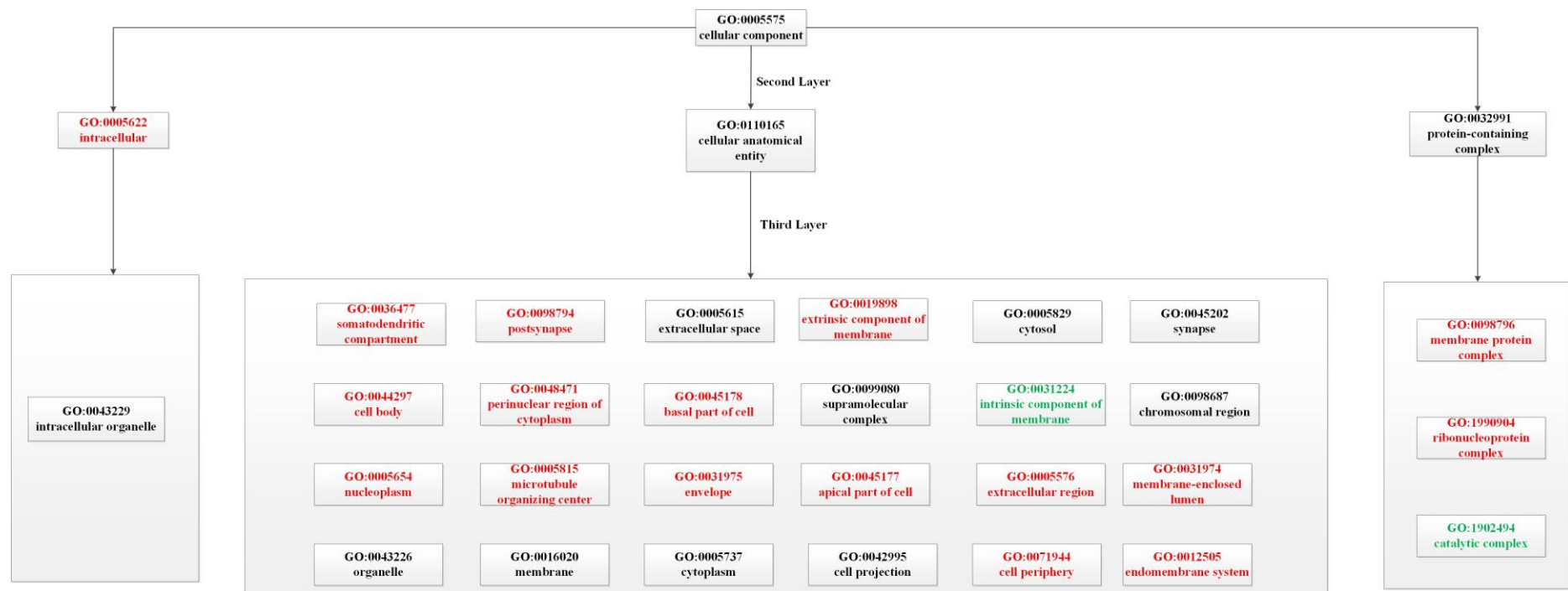
Table S1 Summary of the benchmark dataset

Table S2 Summary of the new dataset

Table S3 Performance of different competing models







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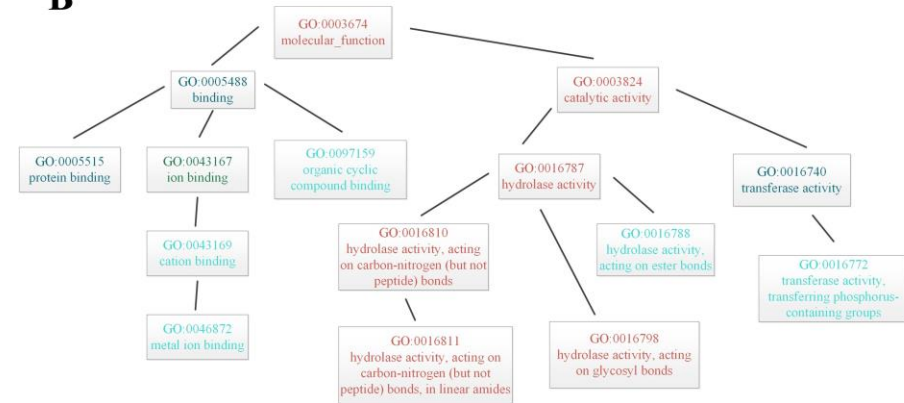
A

Q5QJU3

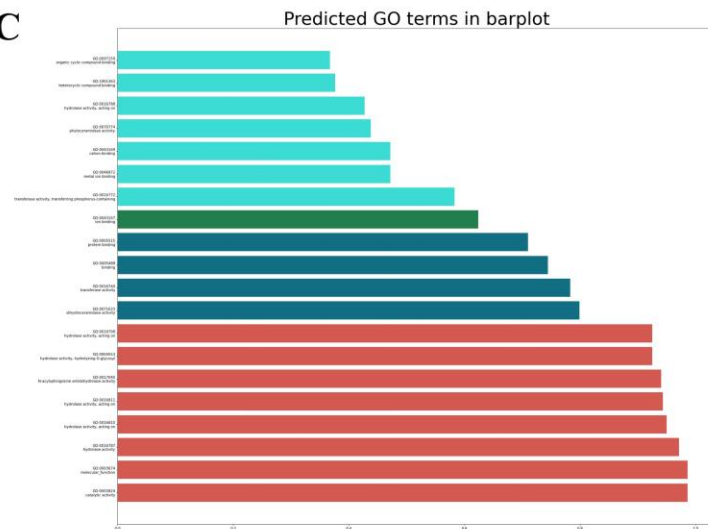
GO DAG Bar plot Bubble plot

GO	Score	BLAST-KNN	LR-InterPro	Net-KNN	LR-ESM	LR-Text	Name	Depth
GO:0003824	0.986	1.0 (3)	0.742 (1)	0.871 (2)	0.977 (2)	0.213 (4)	catalytic activity	2
GO:0003674	0.986	1.0 (3)	0.742 (1)	1.0 (1)	1.0 (1)	1.0 (1)	molecular_function	1
GO:0016787	0.971	1.0 (3)	0.625 (3)	0.436 (4)	0.787 (4)	0.044 (12)	hydrolase activity	3
GO:0016810	0.95	1.0 (3)	0.429 (4)	0.119 (18)	0.787 (4)	--	hydrolase activity, acting on carbon-nitrogen (but not peptide) bonds	4
GO:0016811	0.943	1.0 (3)	0.388 (5)	0.119 (18)	0.787 (4)	--	hydrolase activity, acting on carbon-nitrogen (but not peptide) bonds, in linear amides	5
GO:0017040	0.94	1.0 (3)	0.295 (8)	0.119 (18)	0.787 (4)	--	N-acylsphingosine amidohydrolase activity	6
GO:0004553	0.925	0.766 (7)	0.28 (9)	0.117 (21)	0.508 (8)	--	hydrolase activity, hydrolyzing O-glycosyl compounds	5

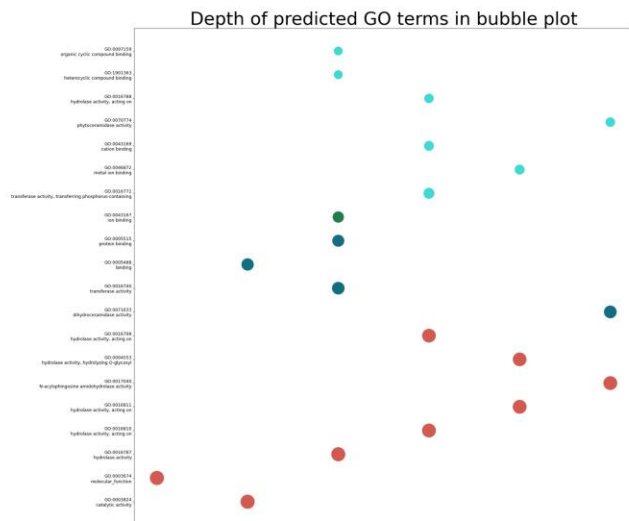
B

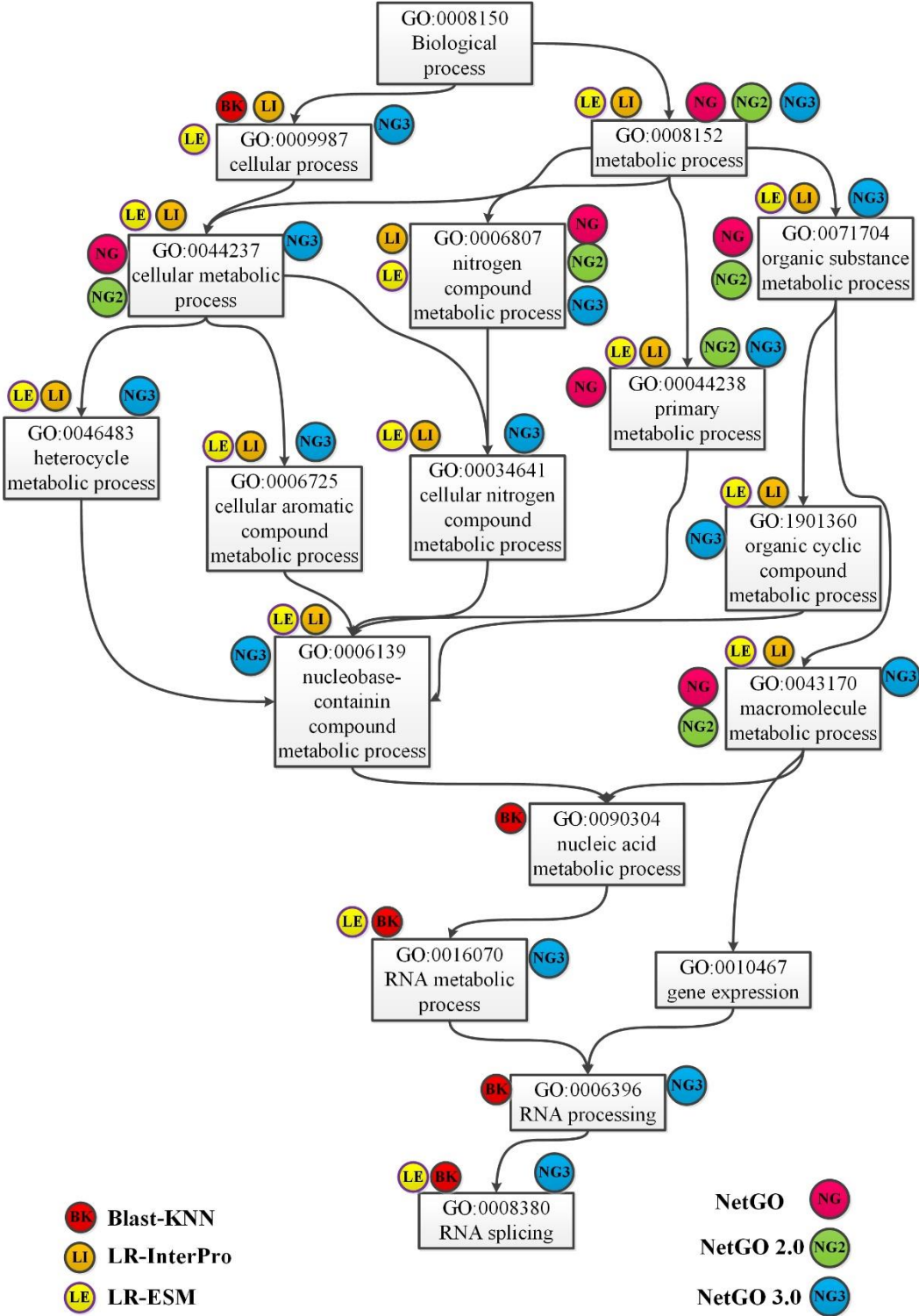


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Table 1 Performance of NetGO 3.0 with its components and competing methods against testing data

	F_{max}			AUPR			S_{min}		
	MFO	BPO	CCO	MFO	BPO	CCO	MFO	BPO	CCO
Naïve	0.416	0.256	0.542	0.276	0.118	0.464	5.683	14.497	6.136
BLAST-KNN	0.632	0.312	0.566	0.542	0.132	0.405	4.098	14.198	5.288
LR-3mer	0.427	0.258	0.552	0.317	0.125	0.478	5.512	14.487	6.035
LR-InterPro	<u>0.651</u>	0.325	<u>0.641</u>	<u>0.623</u>	0.166	<u>0.587</u>	4.055	14.090	5.066
Net-KNN	0.519	0.325	0.596	0.416	0.192	0.528	5.298	13.929	5.554
Seq-RNN	0.524	0.265	0.574	0.424	0.124	0.477	5.129	14.465	5.573
LR-Text	0.464	0.248	0.479	0.353	0.154	0.403	5.362	13.919	5.713
LR-ESM	0.637	<u>0.334</u>	0.631	0.615	<u>0.197</u>	0.572	<u>3.891</u>	<u>13.612</u>	<u>5.052</u>
DeepGOWeb	0.620	0.305	0.620	0.521	0.115	0.493	4.496	14.772	5.550
GOLabeler	0.667	0.326	0.631	0.647	0.193	0.557	3.970	13.558	5.295
NetGO 2.0	0.666	0.366	0.663	0.655	0.269	0.593	4.013	12.984	4.756
NetGO 3.0	0.679	0.378	0.670	0.672	0.268	0.620	3.840	12.800	4.735

470 **Table 2 Performance comparison for proteins in HUMAN**

	F_{max}			AUPR			S_{min}		
	MFO	BPO	CCO	MFO	BPO	CCO	MFO	BPO	CCO
BLAST-KNN	0.655	0.370	0.520	0.581	0.229	0.339	3.186	13.067	4.478
LR-InterPro	<u>0.715</u>	0.373	0.626	<u>0.704</u>	0.307	0.589	<u>3.078</u>	12.732	4.254
Net-KNN	0.598	0.358	0.592	0.565	0.243	0.536	3.882	14.304	4.756
Seq-RNN	0.596	0.291	0.585	0.523	0.184	0.536	3.850	14.729	4.401
LR-ESM	0.711	<u>0.450</u>	<u>0.645</u>	0.690	<u>0.358</u>	<u>0.664</u>	3.105	<u>12.327</u>	<u>3.946</u>
NetGO 2.0	0.715	0.441	0.673	0.725	0.401	0.630	3.018	11.917	3.566
NetGO 3.0	0.721	0.481	0.674	0.730	0.439	0.670	2.929	11.451	3.557

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473 **Table 3 Performance comparison for proteins in MOUSE**

	F_{max}			AUPR			S_{min}		
	MFO	BPO	CCO	MFO	BPO	CCO	MFO	BPO	CCO
BLAST-KNN	0.616	0.353	0.572	0.575	0.147	0.463	5.681	21.804	5.931
LR-InterPro	0.605	0.344	<u>0.591</u>	0.609	<u>0.211</u>	<u>0.542</u>	5.732	21.044	<u>5.489</u>
Net-KNN	0.408	0.341	0.566	0.253	0.199	0.502	8.080	21.309	6.002
Seq-RNN	0.520	0.265	0.537	0.373	0.106	0.462	6.787	22.795	5.993
LR-ESM	<u>0.639</u>	<u>0.352</u>	0.561	<u>0.615</u>	0.197	0.539	<u>5.710</u>	<u>20.639</u>	5.664
NetGO 2.0	0.649	0.420	0.617	0.618	0.315	0.557	5.683	19.572	5.563
NetGO 3.0	0.649	0.427	0.620	0.620	0.316	0.568	5.583	19.545	5.034

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Table 4 Performance comparison comparisons over groups categorized by the number of annotations per GO term

Methods	MFO			BPO			CCO		
	10-30	31-100	>100	10-30	31-100	>100	10-30	31-100	>100
BLAST-KNN	0.628	0.497	0.614	0.197	0.131	0.224	0.265	0.291	0.528
LR-InterPro	0.618	<u>0.562</u>	0.634	0.209	0.138	0.224	0.231	0.307	<u>0.589</u>
Net-KNN	0.330	0.253	0.545	0.139	0.132	0.222	0.210	0.269	0.501
Seq-RNN	0.434	0.326	0.525	0.054	0.062	0.139	0.152	0.195	0.437
LR-ESM	<u>0.642</u>	0.516	<u>0.658</u>	<u>0.307</u>	<u>0.154</u>	<u>0.242</u>	<u>0.333</u>	<u>0.342</u>	0.572
NetGO 2.0	0.658	0.569	0.659	0.248	0.212	0.329	0.300	0.389	0.588
NetGO 3.0	0.675	0.571	0.665	0.250	0.213	0.335	0.386	0.422	0.636

480 **Table 5 Performance on difficult proteins**

	F_{max}			AUPR			S_{min}		
	MFO	BPO	CCO	MFO	BPO	CCO	MFO	BPO	CCO
BLAST-KNN	0.469	0.261	0.386	0.217	0.057	0.206	4.689	13.036	6.128
LR-InterPro	<u>0.614</u>	0.308	0.620	<u>0.551</u>	0.156	0.558	<u>4.018</u>	12.217	5.579
Net-KNN	0.555	0.319	0.591	0.308	0.184	0.528	5.341	12.564	5.543
Seq-RNN	0.486	0.223	0.560	0.312	0.097	0.430	5.264	13.427	5.946
LR-ESM	0.598	<u>0.342</u>	<u>0.631</u>	0.453	<u>0.216</u>	<u>0.594</u>	4.314	<u>12.142</u>	<u>5.271</u>
NetGO 2.0	0.645	0.356	0.634	0.596	0.274	0.595	4.005	11.644	4.888
NetGO 3.0	0.654	0.369	0.668	0.605	0.276	0.609	3.969	11.421	4.782

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483 **Table 6 Performance on proteins with length longer than 1000**

	F_{max}			AUPR			S_{min}		
	MFO	BPO	CCO	MFO	BPO	CCO	MFO	BPO	CCO
BLAST-KNN	0.514	0.272	0.549	0.271	0.119	0.465	6.349	15.176	7.072
LR-InterPro	<u>0.595</u>	0.312	<u>0.638</u>	0.407	0.111	<u>0.603</u>	<u>5.389</u>	<u>15.454</u>	<u>6.253</u>
Net-KNN	0.515	<u>0.329</u>	0.609	<u>0.455</u>	<u>0.205</u>	0.598	6.271	14.510	6.907
Seq-RNN	0.509	0.304	0.587	0.329	0.162	0.508	6.103	14.965	6.903
LR-ESM	0.536	0.309	0.586	0.424	0.135	0.563	5.855	15.213	6.662
NetGO 2.0	0.587	0.357	0.625	0.497	0.241	0.589	5.312	13.824	6.240
NetGO 3.0	0.577	0.348	0.631	0.485	0.215	0.606	5.452	13.947	5.938

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Table 7 Performance on three long proteins in BPO

	AUPR on NetGO 2.0	AUPR on NetGO 3.0	Length
Q3UZV7	0.567	0.471	1028
F1QKQ1	0.472	0.265	1356
Q2HX28	0.404	0.327	1409

489 **Table 8 Predicted GO terms of Q9FGZ9 in BPO by NetGO 3.0 and competing**
 490 **methods**

Method	GO terms			
BLAST-KNN	GO:0009987	GO:0010628	GO:0010604	GO:0010468
	GO:0060255	GO:0009893	GO:0019222	GO:0048518
	GO:0050789	GO:0065007	GO:0000398	GO:0000377
	GO:0000375	GO:0008380	GO:0006397	GO:0016071
	GO:0006396	GO:0016070	GO:0090304	
LR-InterPro	GO:0009987	GO:0008152	GO:0050896	GO:0044237
	GO:0071704	GO:0044238	GO:0006807	GO:0043170
	GO:0065007	GO:0050789	GO:0006950	GO:0044260
	GO:1901564	GO:0034641	GO:0044267	GO:0019538
	GO:1901360	GO:0006725	GO:0046483	
LR-ESM	GO:0009987	GO:0043170	GO:0071704	GO:0008152
	GO:0044238	GO:0006807	GO:0044260	GO:0044237
	GO:0050896	GO:0034641	GO:0046483	GO:1901360
	GO:0006139	GO:0006725	GO:0043412	GO:0008380
	GO:0016070	GO:0000375	GO:000377	
NetGO	GO:0009987	GO:0008152	GO:0044237	GO:0050896
	GO:0043170	GO:0071704	GO:0006807	GO:0044238
	GO:0048856	GO:0032502	GO:0032501	GO:0006950
	GO:0050789	GO:0065007	GO:0007275	GO:0000003
	GO:0022414	GO:0051704	GO:0042221	
NetGO 2.0	GO:0009987	GO:0044237	GO:0008152	GO:0050896
	GO:0071704	GO:0044238	GO:0050789	GO:0065007
	GO:0043170	GO:0006807	GO:0032502	GO:0048856
	GO:0006950	GO:0050794	GO:0007275	GO:0032501
	GO:0003006	GO:0022414	GO:0000003	

NetGO 3.0	GO:0009987	GO:0008152	GO:0044237	GO:0044238
	GO:0043170	GO:0071704	GO:0006807	GO:0050896
	GO:0050789	GO:0065007	GO:1901360	GO:0046483
	GO:0006139	GO:0006725	GO:0034641	GO:0006950
	GO:0008380	GO:0006396	GO:0016070	
Groudtruth	GO:0044237	GO:0044238	GO:0009987	GO:0006139
	GO:0046483	GO:0008150	GO:0006807	GO:0071704
	GO:0006396	GO:0016070	GO:0010467	GO:0008380
	GO:1901360	GO:0008152	GO:0090304	GO:0043170
	GO:0034641	GO:0006725		

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